

MD KAIF PERWEZ

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SUMMARY

Backend-focused Computer Science undergraduate with strong foundations in **Data Structures, Algorithms, OOP, and Distributed Systems fundamentals**. Experienced in designing **scalable APIs, microservices, recommendation systems, and database-optimized backend services** using Node.js, PostgreSQL, Redis, and Python. Seeking **Software Development Engineer** roles.

EDUCATION

Bachelor of Technology in Computer Science Engineering

2022 – 2026

Shri Mata Vaishno Devi University, Jammu

Relevant Coursework: DSA, OOP, DBMS, OS, Computer Networks

SKILLS

Programming	C++, Python, JavaScript, SQL
Backend & Systems	Node.js, Express.js, FastAPI, REST APIs, WebSockets, Microservices
Databases	PostgreSQL, MongoDB, Redis, pgvector, Indexing
Machine Learning	PyTorch, Scikit-learn, Recommendation Systems
Tools	Git, GitHub, Postman, Supabase, Cloudinary
Core CS	Data Structures, Algorithms, OOP, DBMS, Operating Systems

EXPERIENCE

Backend Developer Intern — CodTech IT Solutions

1 Month

- Designed and implemented **RESTful APIs** using Node.js and Express.js following MVC architecture.
- Built a **real-time chat system** using WebSockets enabling low-latency bidirectional communication.
- Developed a user-based **recommendation engine** using interaction history and similarity-based ranking.
- Optimized MongoDB queries using **indexing strategies** improving database performance.

PROJECTS

Xoxia — AI Powered Fashion Recommendation Platform ([GitHub](#))

Built a scalable recommendation platform using **Node.js, Express.js, PostgreSQL, Redis, FastAPI, and PyTorch**. Implemented a hybrid recommendation engine combining **TF-IDF content filtering, SVD collaborative filtering, and Neural Matrix Factorization**. Designed Redis-based event processing, pgvector similarity search, caching, and optimized APIs.

Movie Ticket Booking System ([GitHub](#))

Developed a scalable backend service with **Redis-based distributed seat locking** to handle concurrent booking requests and prevent race conditions. Designed REST APIs using Node.js, Express.js, MongoDB, and Redis.

Notification & Scheduler Service ([GitHub](#))

Built an asynchronous notification microservice with **cron-based scheduling**, email automation, and third-party API integration.

CERTIFICATIONS

Postman API Fundamentals Student Expert

ACHIEVEMENTS

Solved 200+ Data Structures and Algorithms problems on LeetCode.

Participated in multiple national-level software development hackathons.